ISOMORPHISM CLASSES OF FINITE PROJECTIVE MODULES OVER A RING ARE DETECTED MODULO AN IDEAL IN THE JACOBSON RADICAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tags are carried by claim_key and by the external references. The same-chapter pointer lemma-lift-stably-free (0BC5) is counted by a sibling card of this batch; the sentence that refers to it sits at source line 141 and is deliberately kept outside both counted spans.

Allowed standard prerequisites (editorial).

- Nakayama's lemma for a finitely generated module over a ring with an ideal contained in the Jacobson radical. Algebra, Lemma algebra-lemma-NAK (canonical Stacks tag 00DV), invoked at proof lines 162-163. Turns the surjectivity of the two composites modulo I into genuine surjectivity of psi o phi and phi o psi as maps of R-modules.
- A surjective endomorphism of a finite module is an isomorphism. Algebra, Lemma algebra-lemma-fun (canonical Stacks tag 05G8), invoked at proof line 164. Upgrades the surjectivity obtained from Nakayama to invertibility of the two composites, which is what makes phi an isomorphism.
- The lifting property of projective modules against a surjection of rings, applied to finite projective modules. the projectivity of P and P' recorded in the statement at source lines 148-149, used at proof lines 159-161 and 168-170. Produces the lift psi : P' -> P of the composite P' -> P'/IP' -> P/IP and the lift phi : P -> P' of P -> P/IP -> P'/IP'; both are the objects the remainder of the proof operates on.

Source reference. Original lemma 00NZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 877–877. Cross-chapter pointer into the Algebra chapter; it is the standard source of a reduction and the argument built on top of that reduction stays inside the counted proof span.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter pointer into the Algebra

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chapter supplying a standard criterion of finite projectivity; it is background terminology, not part of the counted argument.

Source reference. Original lemma 00DV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 512–512. Cross-chapter pointer into the Algebra chapter for Nakayama's lemma, a standard prerequisite; both applications of it and the surrounding argument stay inside the counted proof span.

Source reference. Original lemma 05G8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7052–7052. Cross-chapter pointer into the Algebra chapter for the fact that a surjective endomorphism of a finite module is an isomorphism, used only to pass from surjectivity to invertibility.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Assume that every element of $1 + I$ is a unit (in other words I is contained in the Jacobson radical of R). If P and P' are finite projective R-modules, then*

(1) *if $\varphi : P \to P'$ is an R-module map inducing an isomorphism $\overline{\varphi} : P/IP \to P'/IP'$, then φ is an isomorphism,*
(2) *if $P/IP \cong P'/IP'$, then $P \cong P'$.*

Proof. Proof of (1). As P' is projective as an R-module we may choose a lift $\psi : P' \to P$ of the map $P' \to P'/IP' \xrightarrow{\overline{\varphi}^{-1}} P/IP$. By Nakayama's lemma (Algebra, Lemma 00DV) $\psi \circ \varphi$ and $\varphi \circ \psi$ are surjective. Hence these maps are isomorphisms (Algebra, Lemma 05G8). Thus φ is an isomorphism.

Proof of (2). Choose an isomorphism $P/IP \cong P'/IP'$. Since P is projective we can choose a lift $\varphi : P \to P'$ of the map $P \to P/IP \to P'/IP'$. Then φ is an isomorphism by (1). □